MACHINE LEARNING PROJECT REPORT

LOAN APPROVAL PREDICTION

Using Logistic Regression

Project Members	Thummar Vrunda Sojitra Krisha
Roll Numbers	525 561
Group No.	5
Subject	Machine Learning
Academic Year	2026–2027

Submitted as a Machine Learning Project Report

1. Certificate

This is to certify that the project report entitled **“Loan Approval Prediction”** has been prepared by **Thummar Vrunda (Roll No. 525)** and **Sojitra Krisha (Roll No. 561)**, Group No. 5, as part of the Machine Learning subject. The report documents the dataset, preprocessing, exploratory analysis, Logistic Regression model building, evaluation, results, conclusion, and future scope of the project.

Student Signature _____ Faculty Signature _____ Date _____

2. Problem Statement

Loan approval is an important decision for financial institutions. Reviewing applications involves considering applicant details such as income, credit score, loan amount, loan term, education, employment status, and asset values. Manual assessment can be time-consuming and may not apply the same decision criteria consistently. The purpose of this project is to develop a machine learning classification model that predicts whether a loan application is likely to be **Approved** or **Rejected** based on applicant information.

The project uses a dataset containing 4,269 applicant records and 13 original fields. The target column is loan_status. A Logistic Regression classifier is trained using the applicant features after categorical encoding and feature scaling.

3. Objectives

• Load and inspect the loan approval dataset.

• Check missing values and duplicate records before model development.

• Explore distributions and relationships among applicant features and loan outcomes.

• Convert categorical values into numerical form.

• Identify and review loan-amount outliers using the IQR method.

• Train a Logistic Regression model for binary loan classification.

• Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.

• Use the trained model to predict loan approval for a new applicant.

4. Dataset Description

The project dataset is stored as **loan_approval_dataset (1).csv**. It contains 4,269 records and 13 original fields. The target variable is loan_status, with two classes: Approved and Rejected.

Loan Status	Applicants	Percentage
Approved	2,656	62.22%
Rejected	1,613	37.78%

Dataset Fields

Field	Description
loan_id	Unique identifier for each applicant record.
no_of_dependents	Number of dependents.
education	Graduate or Not Graduate.
self_employed	Whether the applicant is self-employed.
income_annum	Applicant's annual income.
loan_amount	Requested loan amount.
loan_term	Loan term; unit should be confirmed from dataset documentation.
cibil_score	Applicant credit score.
residential_assets_value	Value of residential assets.
commercial_assets_value	Value of commercial assets.
luxury_assets_value	Value of luxury assets.
bank_asset_value	Value of bank assets.
loan_status	Target class: Approved or Rejected.

5. Data Preprocessing

Data preprocessing prepares the applicant data for machine learning. The supplied Python code removes duplicate rows, standardizes categorical text, converts categories into numeric values, filters loan-amount outliers using the IQR rule, and scales numeric features before model training.

Data cleaning and transformation

• Remove extra spaces from column names using df.columns.str.strip().

• Remove duplicate records using drop_duplicates().

• Trim and convert education, self_employed, and loan_status values to lowercase.

• Encode education: Graduate = 1 and Not Graduate = 0.

• Encode self_employed: Yes = 1 and No = 0.

• Encode loan_status: Approved = 1 and Rejected = 0.

• Remove loan_id because it is an identifier rather than a predictive applicant feature.

• Apply StandardScaler to the training features and transform the test features using the fitted scaler.

Outlier handling

The code calculates the first quartile (Q1), third quartile (Q3), and interquartile range (IQR) for loan_amount. Values outside Q1 – 1.5 × IQR and Q3 + 1.5 × IQR are filtered out. For this dataset, the standard IQR rule flags no loan-amount outliers, so all 4,269 records remain after this step.

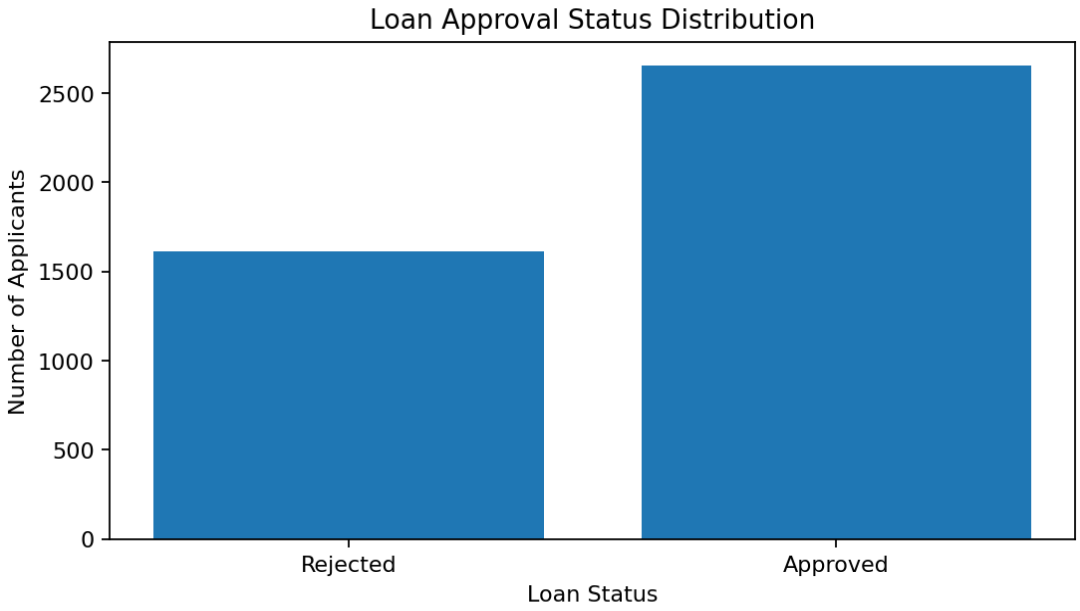
Data quality observations

The CSV contains no missing values and no duplicate rows. However, 28 records have a negative residential_assets_value; these values should be checked against the original data source before any correction is made.

6. Exploratory Data Analysis

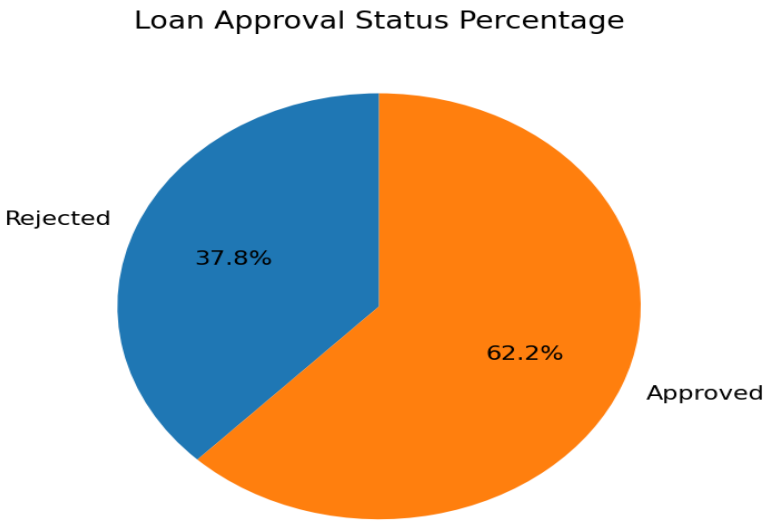
EDA is used to understand the target-class distribution and applicant feature patterns before model training.

Loan approval status distribution



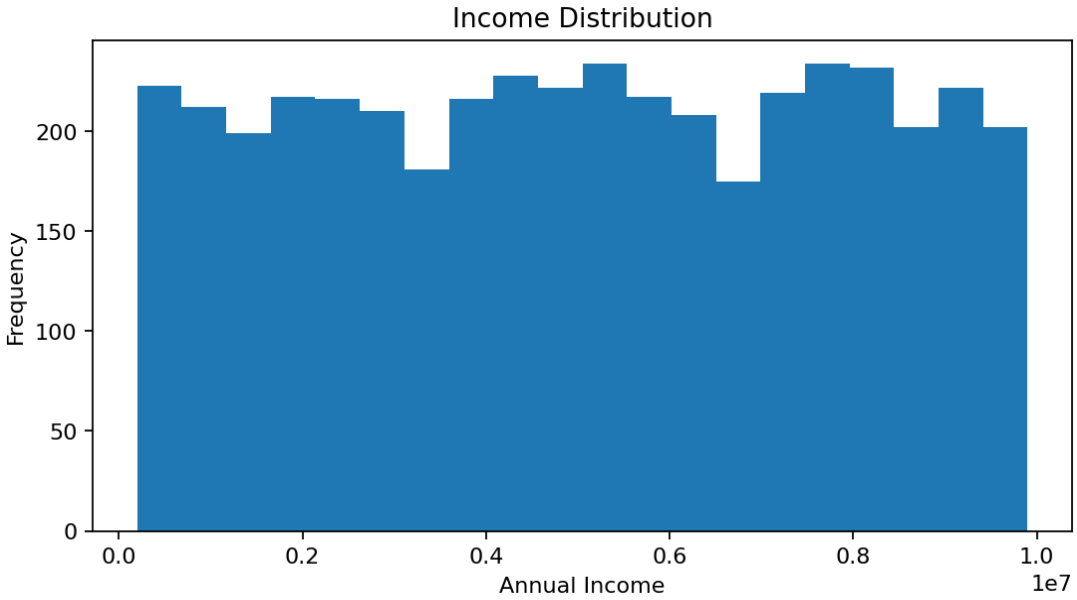
The dataset contains 2,656 approved applications (62.22%) and 1,613 rejected applications (37.78%). The two classes are not equal in size, but both contain substantial numbers of records.

Loan status percentage



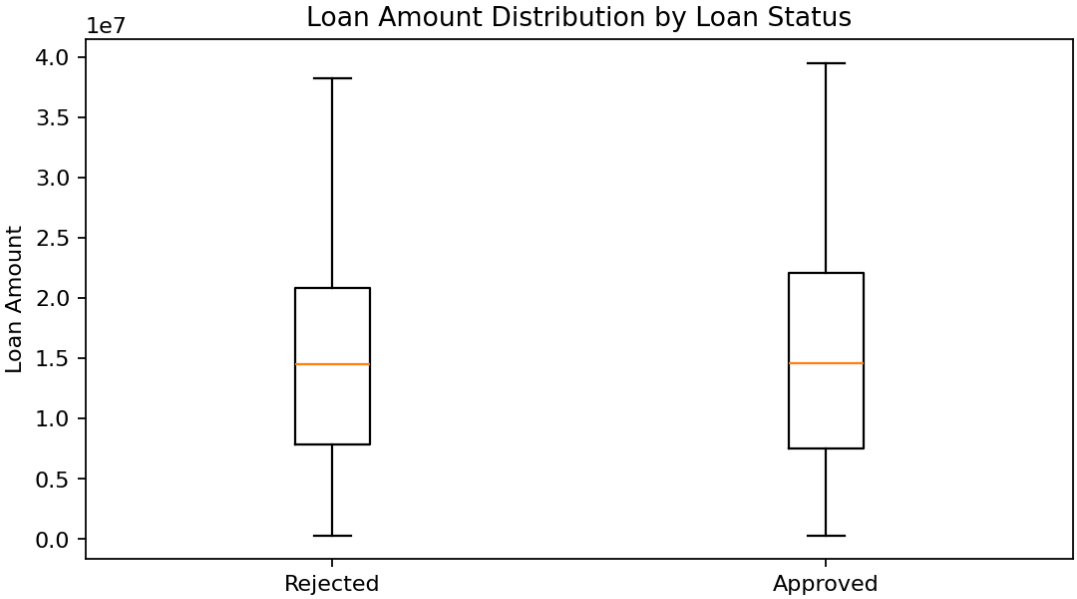
6. Exploratory Data Analysis (continued)

Income distribution



Annual income ranges from 200,000 to 9,900,000 in the dataset's original units. The mean annual income is 5,059,124.

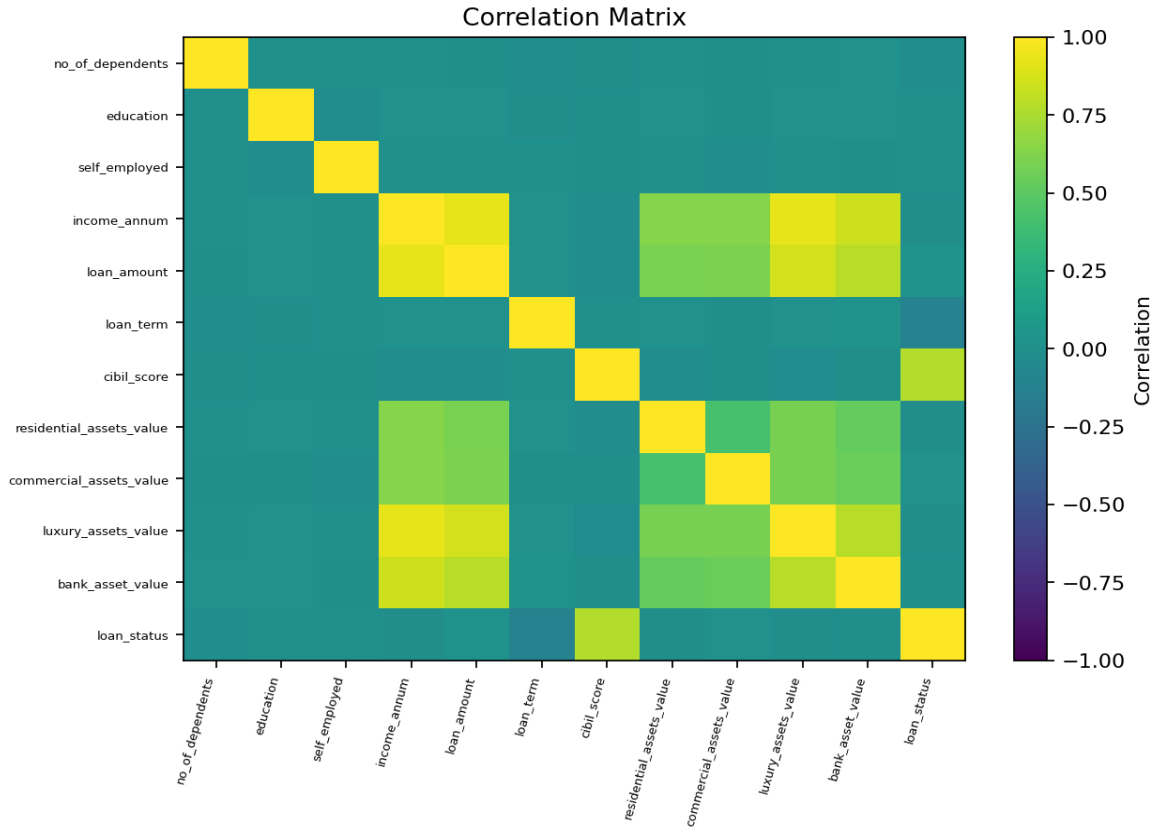
Loan amount distribution by loan status



The box plot compares requested loan amounts for approved and rejected applications. The distributions overlap; this feature should be considered together with credit score, income, loan term, and asset information.

6. Exploratory Data Analysis (continued)

Correlation matrix



The correlation matrix summarizes pairwise linear relationships among the numeric and encoded variables. Correlation indicates association and does not establish causation.

7. Model Building (Code)

The supplied implementation uses Logistic Regression. The loan_id column is excluded, the remaining applicant attributes are used as features, and loan_status is the target.

Train/test split and feature scaling

The dataset is divided using an 80/20 stratified split with random_state=42. This results in **3,415 training records** and **854 testing records**. StandardScaler is fitted on the training features and then applied to the test features.

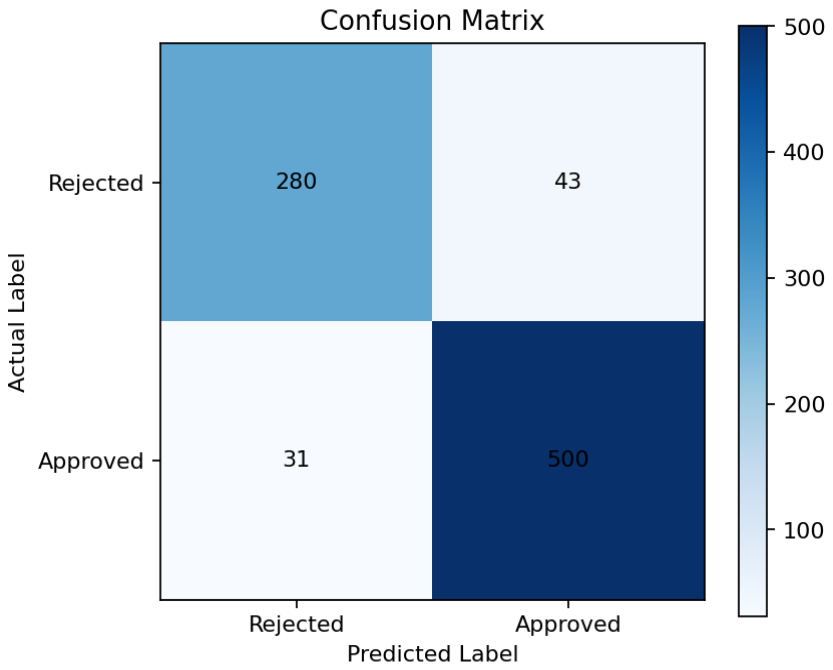
Model training

LogisticRegression(max_iter=1000, random_state=42) is fitted on the scaled training data. The trained model predicts Approved or Rejected for the held-out test set.

8. Model Evaluation

The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. Approved (class 1) is treated as the positive class.

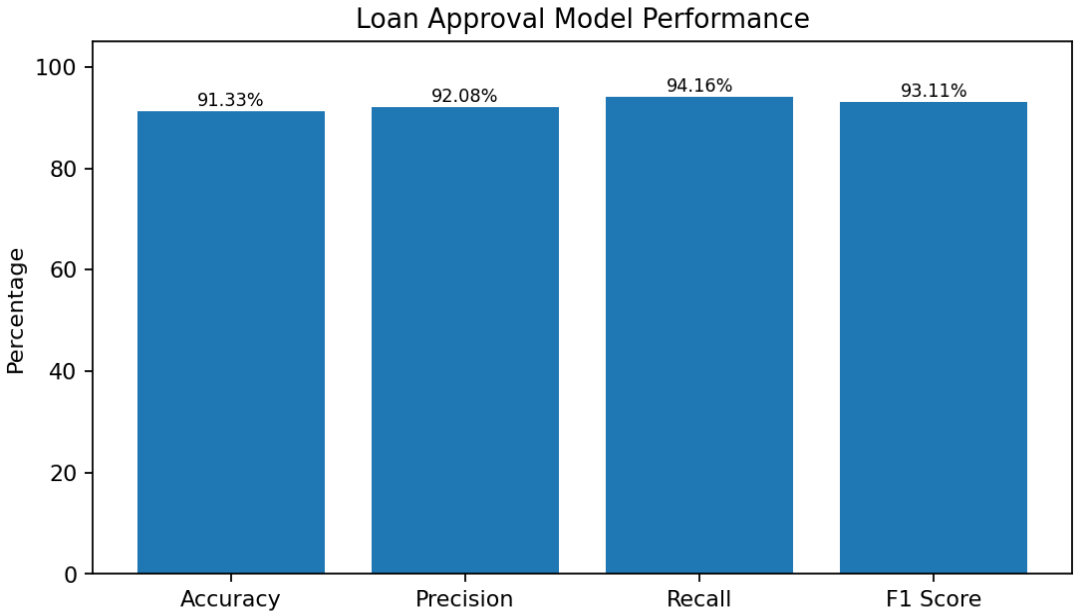
Metric	Value	Meaning
Accuracy	91.33%	Overall proportion of correct predictions.
Precision	92.08%	Among predicted approvals, proportion that were actually approved.
Recall	94.16%	Among actual approvals, proportion correctly identified.
F1 Score	93.11%	Harmonic mean of precision and recall.



The confusion matrix contains 280 correctly predicted rejected applications, 43 rejected applications predicted as approved, 31 approved applications predicted as rejected, and 500 correctly predicted approved applications.

9. Results

On the held-out test set of 854 records, the Logistic Regression model achieved the following results:

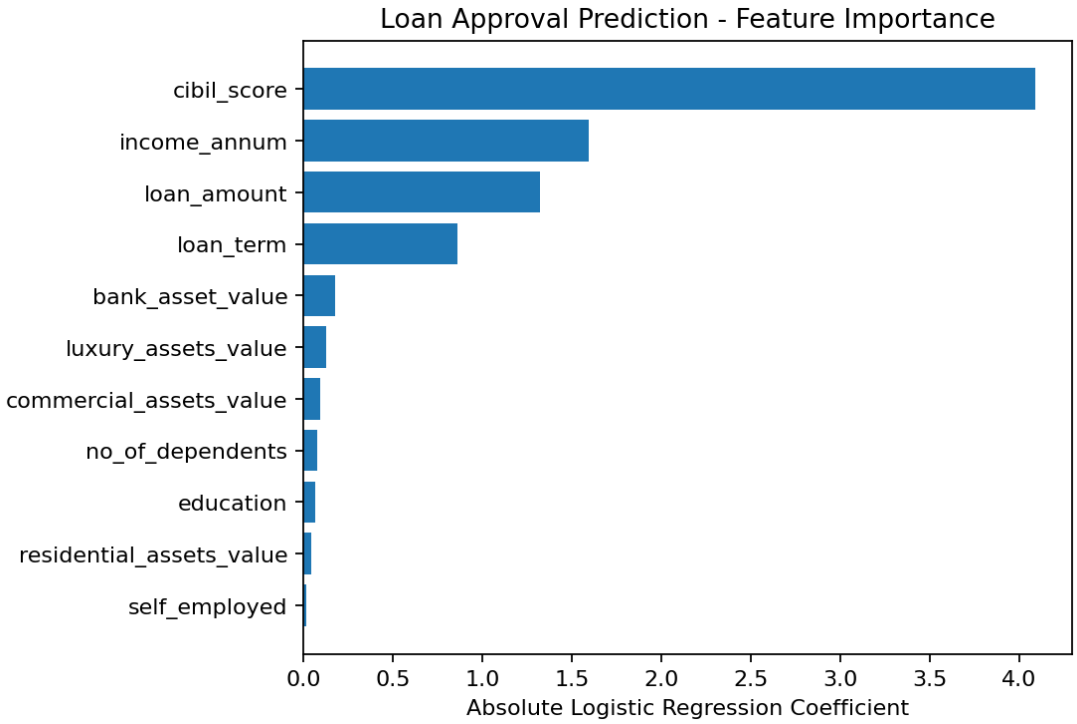


Evaluation Metric	Result
Accuracy	91.33%
Precision	92.08%
Recall	94.16%
F1 Score	93.11%

5-Fold Cross Validation

The supplied code applies 5-fold cross-validation to the scaled training features. The fold accuracies are: 92.24%, 90.92%, 90.92%, 91.95%, 91.22%. The average cross-validation accuracy is **91.45%**.

Feature importance



The feature chart displays the absolute values of Logistic Regression coefficients after scaling. Absolute values indicate coefficient magnitude, not whether a feature increases or decreases the likelihood of approval.

10. Conclusion

This project demonstrates a machine learning workflow for loan approval prediction using applicant information. It includes dataset inspection, preprocessing, exploratory analysis, categorical encoding, feature scaling, Logistic Regression training, evaluation, and prediction. With the supplied dataset and an 80/20 stratified split, the reproduced workflow achieved **91.33% accuracy**, **92.08% precision**, **94.16% recall**, and an **F1-score of 93.11%** on the held-out test set. These results describe this dataset and split and do not guarantee the same performance on unseen real-world applications.

11. Future Scope

• Test the model on larger and more diverse loan application datasets.

• Compare Logistic Regression with Decision Tree, Random Forest, Support Vector Machine, and other classification algorithms.

• Perform hyperparameter tuning and use leakage-safe cross-validation pipelines.

• Investigate negative residential asset values and confirm the units of all monetary and loan-term fields.

• Build a simple web interface where users can enter applicant details and receive a prediction.

• Review model fairness, explainability, and performance across relevant applicant groups before real-world use.

12. Project Files and Implementation Notes

File / Item	Purpose
All Code.py	Python implementation of the Loan Approval Prediction workflow.
loan_approval_dataset (1).csv	Applicant dataset containing 4,269 records and 13 fields.
loan_approval_model.pkl	Saved Logistic Regression model included in the ZIP.
loan_scaler.pkl	Saved StandardScaler included in the ZIP.
Images/*.png	Loan status, pie chart, loan amount, confusion matrix, model performance, and feature importance graphs.

Code review note

The supplied script contains a few naming inconsistencies: it refers to income_amount in the income histogram and new-applicant sample, while the uploaded CSV column is income_annum. The model metrics in this report were reproduced using the actual CSV column name. The script also performs scaling before cross-validation; a Pipeline with StandardScaler and LogisticRegression would provide a more reliable cross-validation estimate by fitting scaling within each fold.

— End of Report —